

Practical Examination

Deploy a Student Event Registration Portal

Problem Statement

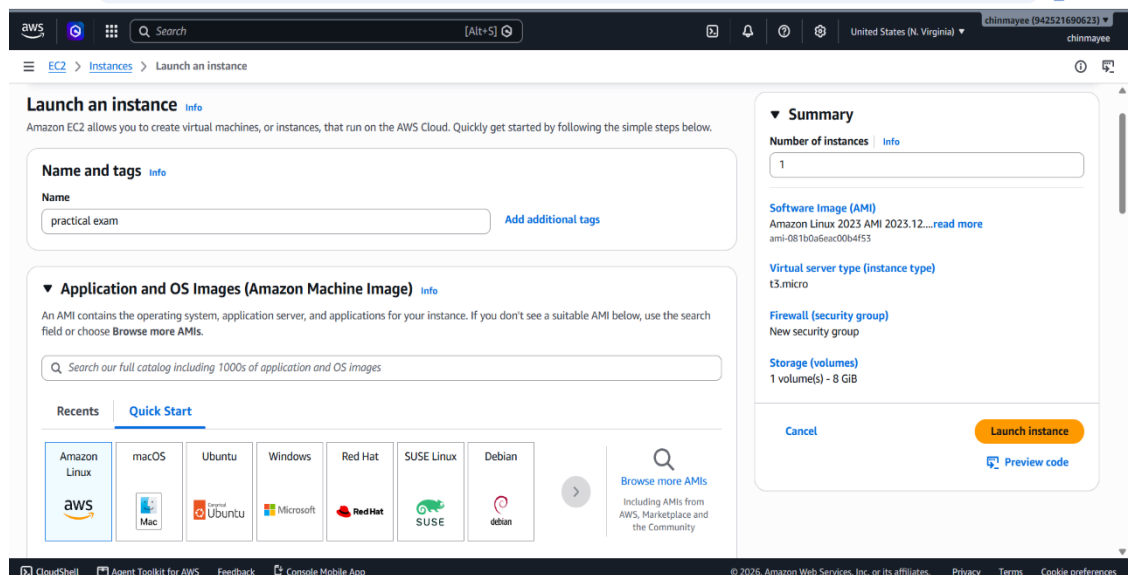
You have been assigned to deploy a **Student Event Registration Portal** on a Linux server.

You must build and host a simple dynamic website that allows students to **register for an event and log in using their registered credentials**.

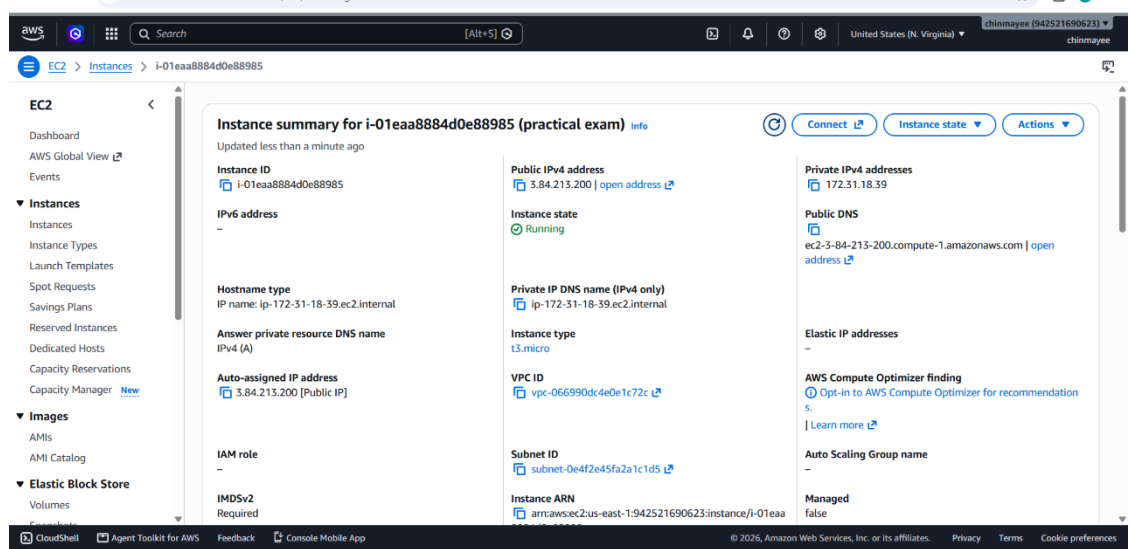
The website must be deployed using **either LAMP or LEMP** and must use **MySQL/MariaDB** for storing user information.

1. AWS EC2

- EC2 instance creation



- Instance running



AWS

[Alt+S] Search United States (N. Virginia) chinmayee (942521690623) chinmayee

EC2 > Instances > Launch an instance

Network settings Info Edit

Network Info
vpc-066990dc4e0e1c72c

Subnet Info
No preference (Default subnet in any availability zone)

Auto-assign public IP Info
Enable

Firewall (security groups) Info
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

- Create security group Select existing security group

We'll create a new security group called 'launch-wizard-4' with the following rules:

Action	Description	CIDR/IP range
☒	Allow SSH traffic from Helps you connect to your instance	Anywhere 0.0.0.0/0
☐	Allow HTTPS traffic from the internet To set up an endpoint, for example when creating a web server	
☒	Allow HTTP traffic from the internet To set up an endpoint, for example when creating a web server	

Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Summary

Number of instances Info
1

Software Image (AMI)
Amazon Linux 2023 AMI 2023.12...read more
ami-0B1BCaBeac00b4f53

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

Cancel Launch instance Preview code

- SSH connection
- Linux user creation
- Important Linux commands/results

[illegible]

3. LAMP/LEMP

- Installation

```
ec2-user@ip-172-31-18-39: ~]$ sudo yum update -y
Amazon Linux 2023 Kernel Livepatch repository
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-172-31-18-39 ~]$ sudo yum install nginx mariadb105-server php -y
Last metadata expiration check: 0:01:12 ago on Tue Sep  8 14:36:16 2026.
Dependencies resolved.
=====
Package                                Architecture      Version           Repository        Size
=====
Installing:
mariadb105-server                      x86_64            3:10.5.29-1.amzn2023.0.1  amazonlinux      10 M
nginx                                   x86_64            1:1.30.4-1.amzn2023.0.1  amazonlinux       34 k
php8.5                                  x86_64            8.5.9-1.amzn2023.0.1    amazonlinux       14 k
Installing dependencies:
apr                                      x86_64            1.7.5-1.amzn2023.0.4    amazonlinux       129 k
apr-util                               x86_64            1.6.5-1.amzn2023.0.1    amazonlinux       100 k
apr-util-ldap                          x86_64            1.6.5-1.amzn2023.0.1    amazonlinux       15 k
generic-logos-httpd                   noarch            18.0.0-12.amzn2023.0.3  amazonlinux       19 k
gperftools-libs                       x86_64            2.9.1-1.amzn2023.0.3    amazonlinux       388 k
httpd-core                             x86_64            2.4.68-1.amzn2023.0.1    amazonlinux       1.4 M
httpd-filesystem                      noarch            2.4.68-1.amzn2023.0.1    amazonlinux       12 k
httpd-tools                           x86_64            2.4.68-1.amzn2023.0.1    amazonlinux       80 k
libbrotli                             x86_64            1.0.9-4.amzn2023.0.2    amazonlinux       315 k
libsodium                             x86_64            1.0.19-5.amzn2023.0.1    amazonlinux       174 k
libunwind                             x86_64            1.4.0-5.amzn2023.0.3    amazonlinux       66 k
mailcap                               noarch            2.1.49-3.amzn2023.0.3    amazonlinux       33 k
mariadb-connector-c                   x86_64            3.3.10-1.amzn2023.0.2    amazonlinux       211 k
mariadb-connector-c-config            x86_64            3.3.10-1.amzn2023.0.2    amazonlinux       9.3 k
mariadb105                            x86_64            3:10.5.29-1.amzn2023.0.1  amazonlinux       1.5 M
mariadb105-common                    x86_64            3:10.5.29-1.amzn2023.0.1  amazonlinux       28 k
mariadb105-errmsg                     x86_64            3:10.5.29-1.amzn2023.0.1  amazonlinux       212 k
mysql-selinux                         noarch            1.0.14-2.amzn2023.0.1    amazonlinux       36 k
nginx-core                            x86_64            1:1.30.4-1.amzn2023.0.1    amazonlinux       711 k
nginx-filesystem                      noarch            1:1.30.4-1.amzn2023.0.1    amazonlinux       10 k
nginx-mimetypes                      noarch            2.1.49-3.amzn2023.0.3    amazonlinux       21 k
perl-DBD-MariaDB                     x86_64            1.22-1.amzn2023.0.5      amazonlinux       152 k
perl-DBI                             x86_64            1.652-1.amzn2023.0.1     amazonlinux       811 k
perl-File-Copy                       noarch            2.34-477.amzn2023.0.9    amazonlinux       20 k
perl-Math-BigInt                     noarch            1:1.9998.39-2.amzn2023.0.2  amazonlinux       282 k
=====
```

○Nginx status

```
ec2-user@ip-172-31-18-39: ~]$ sudo service nginx start
Redirecting to /bin/systemctl start nginx.service
[ec2-user@ip-172-31-18-39 ~]$ sudo systemctl enable nginx
Created symlink /etc/systemd/system/multi-user.target.wants/nginx.service → /usr/lib/systemd/system/nginx.service.
[ec2-user@ip-172-31-18-39 ~]$ sudo service nginx status
Redirecting to /bin/systemctl status nginx.service
● nginx.service - The nginx HTTP and reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: disabled)
   Drop-In: /usr/lib/systemd/system/nginx.service.d
            └─php-fpm.conf
   Active: active (running) since Tue 2026-09-08 14:39:39 UTC; 45s ago
     Main PID: 28149 (nginx)
       Tasks: 3 (limit: 1059)
      Memory: 3.5M
         CPU: 80ms
    CGroup: /system.slice/nginx.service
            └─28149 "nginx: master process /usr/sbin/nginx"
              └─28150 "nginx: worker process"
                └─28153 "nginx: worker process"

Sep 08 14:39:39 ip-172-31-18-39.ec2.internal systemd[1]: Starting nginx.service - The nginx HTTP and reverse proxy server...
Sep 08 14:39:39 ip-172-31-18-39.ec2.internal nginx[28144]: nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
Sep 08 14:39:39 ip-172-31-18-39.ec2.internal nginx[28144]: nginx: configuration file /etc/nginx/nginx.conf test is successful
Sep 08 14:39:39 ip-172-31-18-39.ec2.internal systemd[1]: Started nginx.service - The nginx HTTP and reverse proxy server.
[ec2-user@ip-172-31-18-39 ~]$ nginx -v
nginx version: nginx/1.30.4
[ec2-user@ip-172-31-18-39 ~]$
```

○ MySQL/MariaDB status

```
ec2-user@ip-172-31-18-39 ~]$ sudo systemctl start mariadb
[ec2-user@ip-172-31-18-39 ~]$ sudo systemctl enable mariadb
Created symlink /etc/systemd/system/mysql.service → /usr/lib/systemd/system/mariadb.service.
Created symlink /etc/systemd/system/mysqld.service → /usr/lib/systemd/system/mariadb.service.
Created symlink /etc/systemd/system/multi-user.target.wants/mariadb.service → /usr/lib/systemd/system/mariadb.service.
[ec2-user@ip-172-31-18-39 ~]$ sudo service mariadb status
Redirecting to /bin/systemctl status mariadb.service
● mariadb.service - MariaDB 10.5 database server
   Loaded: loaded (/usr/lib/systemd/system/mariadb.service; enabled; preset: disabled)
   Active: active (running) since Tue 2026-09-08 14:44:51 UTC; 1min 57s ago
     Docs: man:mariadb(8)
           https://mariadb.com/kb/en/library/systemd/
  Main PID: 28658 (mariadb)
    Status: "Taking your SQL requests now..."
     Tasks: 8 (limit: 1059)
  Memory: 66.1M
     CPU: 519ms
  CGroup: /system.slice/mariadb.service
          └─28658 /usr/libexec/mariadb --basedir=/usr

Sep 08 14:44:51 ip-172-31-18-39.ec2.internal mariadb-prepare-db-dir[28613]: The second is mysql@localhost, it has no password either, but
Sep 08 14:44:51 ip-172-31-18-39.ec2.internal mariadb-prepare-db-dir[28613]: you need to be the system 'mysql' user to connect.
Sep 08 14:44:51 ip-172-31-18-39.ec2.internal mariadb-prepare-db-dir[28613]: After connecting you can set the password, if you would need to be
Sep 08 14:44:51 ip-172-31-18-39.ec2.internal mariadb-prepare-db-dir[28613]: able to connect as any of these users with a password and without sudo
Sep 08 14:44:51 ip-172-31-18-39.ec2.internal mariadb-prepare-db-dir[28613]: See the MariaDB Knowledgebase at https://mariadb.com/kb
Sep 08 14:44:51 ip-172-31-18-39.ec2.internal mariadb-prepare-db-dir[28613]: Please report any problems at https://mariadb.org/jira
Sep 08 14:44:51 ip-172-31-18-39.ec2.internal mariadb-prepare-db-dir[28613]: The latest information about MariaDB is available at https://mariadb.org/.
Sep 08 14:44:51 ip-172-31-18-39.ec2.internal mariadb-prepare-db-dir[28613]: Consider joining MariaDB's strong and vibrant community:
Sep 08 14:44:51 ip-172-31-18-39.ec2.internal mariadb-prepare-db-dir[28613]: https://mariadb.org/get-involved/
Sep 08 14:44:51 ip-172-31-18-39.ec2.internal systemd[1]: Started mariadb.service - MariaDB 10.5 database server.
[ec2-user@ip-172-31-18-39 ~]$
```

○ PHP installation/version

```
ec2-user@ip-172-31-18-39 ~]$ php -v
PHP 8.5.9 (cli) (built: Jul 28 2026 13:06:52) (NTS gcc x86_64)
Copyright (c) The PHP Group
Built by Amazon Linux
Zend Engine v4.5.9, Copyright (c) Zend Technologies
    with Zend OPcache v8.5.9, Copyright (c), by Zend Technologies
[ec2-user@ip-172-31-18-39 ~]$ sudo service php-fpm start
Redirecting to /bin/systemctl start php-fpm.service
[ec2-user@ip-172-31-18-39 ~]$ sudo systemctl enable php-fpm
Created symlink /etc/systemd/system/multi-user.target.wants/php-fpm.service → /usr/lib/systemd/system/php-fpm.service.
[ec2-user@ip-172-31-18-39 ~]$ sudo service php-fpm status
Redirecting to /bin/systemctl status php-fpm.service
● php-fpm.service - The PHP FastCGI Process Manager
   Loaded: loaded (/usr/lib/systemd/system/php-fpm.service; enabled; preset: disabled)
   Active: active (running) since Tue 2026-09-08 14:39:39 UTC; 3min 53s ago
     Main PID: 28143 (php-fpm)
    Status: "Processes active: 0, idle: 5, Requests: 0, slow: 0, Traffic: 0.00req/sec"
     Tasks: 6 (limit: 1059)
  Memory: 11.1M
     CPU: 79ms
  CGroup: /system.slice/php-fpm.service
          └─28143 "php-fpm: master process (/etc/php-fpm.conf)"
              └─28146 "php-fpm: pool www"
                  └─28147 "php-fpm: pool www"
                      └─28148 "php-fpm: pool www"
                          └─28151 "php-fpm: pool www"
                              └─28152 "php-fpm: pool www"

Sep 08 14:39:39 ip-172-31-18-39.ec2.internal systemd[1]: Starting php-fpm.service - The PHP FastCGI Process Manager...
Sep 08 14:39:39 ip-172-31-18-39.ec2.internal systemd[1]: Started php-fpm.service - The PHP FastCGI Process Manager.
[ec2-user@ip-172-31-18-39 ~]$
```

4. SQL

- Database creation ○ Table creation

```
ec2-user@ip-172-31-18-39 ~]$ sudo mariadb -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 3
Server version: 10.5.29-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> CREATE DATABASE campusconnect;
Query OK, 1 row affected (0.000 sec)

MariaDB [(none)]> USE campusconnect;
Database changed
MariaDB [campusconnect]> CREATE TABLE students (
->   id INT AUTO_INCREMENT PRIMARY KEY,
->   full_name VARCHAR(100) NOT NULL,
->   student_id VARCHAR(50) NOT NULL UNIQUE,
->   email VARCHAR(100) NOT NULL UNIQUE,
->   college_name VARCHAR(150) NOT NULL,
->   location VARCHAR(100) NOT NULL,
->   event_name VARCHAR(100) NOT NULL,
->   password VARCHAR(255) NOT NULL,
->   created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
-> );
Query OK, 0 rows affected (0.012 sec)

MariaDB [campusconnect]> SHOW TABLES;
+-----+
| Tables_in_campusconnect |
+-----+
| students                 |
+-----+
1 row in set (0.000 sec)

MariaDB [campusconnect]> DESC students;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| id    | int(11) | NO   | PRI | NULL    | auto_increment |
+-----+-----+-----+-----+-----+-----+
```

```
ec2-user@ip-172-31-18-39 ~]$ sudo mariadb -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 3
Server version: 10.5.29-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> CREATE DATABASE campusconnect;
Query OK, 1 row affected (0.000 sec)

MariaDB [(none)]> USE campusconnect;
Database changed
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->   id INT AUTO_INCREMENT PRIMARY KEY,
->   full_name VARCHAR(100) NOT NULL,
->   student_id VARCHAR(50) NOT NULL UNIQUE,
->   email VARCHAR(100) NOT NULL UNIQUE,
->   college_name VARCHAR(150) NOT NULL,
->   location VARCHAR(100) NOT NULL,
->   event_name VARCHAR(100) NOT NULL,
->   password VARCHAR(255) NOT NULL,
->   created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
-> );
Query OK, 0 rows affected (0.012 sec)

MariaDB [campusconnect]> SHOW TABLES;
+-----+
| Tables_in_campusconnect |
+-----+
| students                 |
+-----+
1 row in set (0.000 sec)

MariaDB [campusconnect]> DESC students;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| id    | int(11) | NO   | PRI | NULL    | auto_increment |
| full_name | varchar(100) | NO   |     | NULL    |               |
| student_id | varchar(50) | NO   | UNI | NULL    |               |
| email    | varchar(100) | NO   | UNI | NULL    |               |
| college_name | varchar(150) | NO   |     | NULL    |               |
| location | varchar(100) | NO   |     | NULL    |               |
| event_name | varchar(100) | NO   |     | NULL    |               |
| password | varchar(255) | NO   |     | NULL    |               |
| created_at | timestamp | NO   |     | current_timestamp() |           |
+-----+-----+-----+-----+-----+-----+
9 rows in set (0.001 sec)

MariaDB [campusconnect]> exit
Bye
ec2-user@ip-172-31-18-39 ~]$
```

- SQL queries ○ Inserted registration data

```
aws [Search] [Alt+S] Ask Amazon Q United States (N. Virginia) chinmayee (942521690623) chinmayee

[ec2-user@ip-172-31-18-39 ~]$ mysql -u campususer -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 14
Server version: 10.5.29-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> USE campusconnect;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
MariaDB [campusconnect]> SHOW TABLES;
+-----+
| Tables in campusconnect |
+-----+
| students                |
+-----+
1 row in set (0.000 sec)

MariaDB [campusconnect]> desc students;
+-----+-----+-----+-----+-----+-----+
| Field      | Type      | Null | Key | Default | Extra      |
+-----+-----+-----+-----+-----+-----+
| id         | int(11)   | NO   | PRI | NULL    | auto_increment |
| full_name  | varchar(100) | NO   |     | NULL    |              |
| student_id | varchar(50) | NO   | UNI | NULL    |              |
| email      | varchar(100) | NO   | UNI | NULL    |              |
| college_name | varchar(150) | NO   |     | NULL    |              |
+-----+-----+-----+-----+-----+-----+

i-01eaa8884d0e88985 (practical exam)
PublicIPs: 3.84.213.200 PrivateIPs: 172.31.18.39
```

- SELECT output showing registered students.

```
aws [Search] [Alt+S] Ask Amazon Q United States (N. Virginia) chinmayee (942521690623) chinmayee

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> USE campusconnect;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
MariaDB [campusconnect]> SHOW TABLES;
+-----+
| Tables in campusconnect |
+-----+
| students                |
+-----+
1 row in set (0.000 sec)

MariaDB [campusconnect]> SELECT * FROM students;
Empty set (0.000 sec)

MariaDB [campusconnect]> SELECT * FROM students;
+-----+-----+-----+-----+-----+-----+-----+-----+
| id | full_name | student_id | email      | college_name | location | event_name      | password      | created_at      |
+-----+-----+-----+-----+-----+-----+-----+-----+
| 7  | naina     | 1          | g@34.com   | DY patil     | pune     | Cultural Event  | $2y$12$MccoICDgTDGhwxCNWT5P0.NX1ThkigqPaHoZcpJ0oyTStlhjQ7Ufm | 2026-09-08 18:59:08 |
+-----+-----+-----+-----+-----+-----+-----+-----+
1 row in set (0.000 sec)

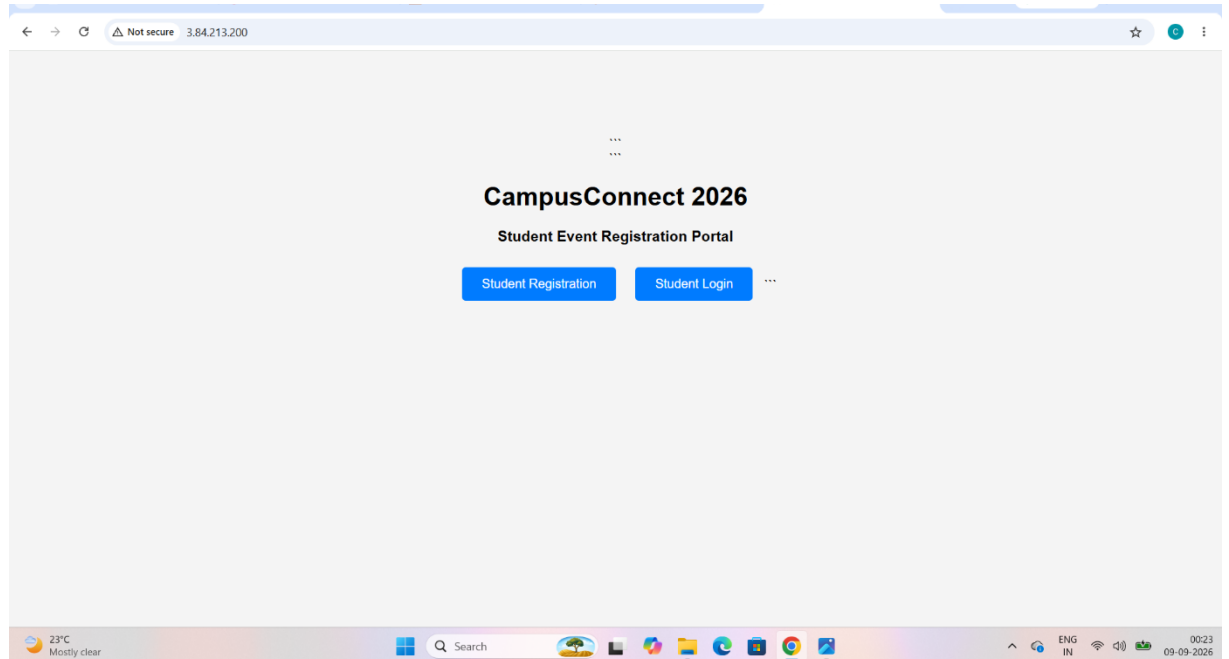
MariaDB [campusconnect]>
```

○ Create Database connection files

```
ec2-user@ip-172-31-18-39: /usr/share/nginx/html
[ec2-user@ip-172-31-18-39 ~]$ cd /usr/share/nginx/html
[ec2-user@ip-172-31-18-39 html]$ ls -la
total 16
drwxr-xr-x. 3 root root 112 Sep  8 14:37 .
drwxr-xr-x. 4 root root 33 Sep  8 14:37 ..
-rw-r--r--. 1 root root 3464 Jul 20 22:28 404.html
-rw-r--r--. 1 root root 3507 Jul 20 22:28 50x.html
drwxr-xr-x. 2 root root 27 Sep  8 14:37 icons
-rw-r--r--. 1 root root 896 Jul 20 22:30 index.html
-rw-r--r--. 1 root root 368 Jul 20 22:28 nginx-logo.png
lrwxrwxrwx. 1 root root 14 Jul 20 22:30 poweredby.png -> nginx-logo.png
[ec2-user@ip-172-31-18-39 html]$ sudo rm -f index.html
[ec2-user@ip-172-31-18-39 html]$ sudo touch index.php register.php signup.php login.php authenticate.php
[ec2-user@ip-172-31-18-39 html]$ sudo chown -R ec2-user:ec2-user /usr/share/nginx/html
[ec2-user@ip-172-31-18-39 html]$ nano /usr/share/nginx/html/config.php
[ec2-user@ip-172-31-18-39 html]$ nano /usr/share/nginx/html/index.php
[ec2-user@ip-172-31-18-39 html]$ nano /usr/share/nginx/html/register.php
[ec2-user@ip-172-31-18-39 html]$ nano /usr/share/nginx/html/signup.php
[ec2-user@ip-172-31-18-39 html]$ nano /usr/share/nginx/html/signup.php
[ec2-user@ip-172-31-18-39 html]$ nano /usr/share/nginx/html/login.php
[ec2-user@ip-172-31-18-39 html]$ nano /usr/share/nginx/html/authenticate.phpnano /usr/share/nginx/html/authenticate.phpnano /usr/share/nginx/html/authenticate.php
[ec2-user@ip-172-31-18-39 html]$ ls
404.html authenticate.php config.php index.php nginx-logo.png register.php
50x.html authenticate.phpnano icons login.php poweredby.png signup.php
[ec2-user@ip-172-31-18-39 html]$ rm -r authenticate.phpnano
[ec2-user@ip-172-31-18-39 html]$ nano /usr/share/nginx/html/authenticate.php
[ec2-user@ip-172-31-18-39 html]$ sudo nano /etc/nginx/nginx.conf
[ec2-user@ip-172-31-18-39 html]$ sudo nano /etc/nginx/nginx.conf
[ec2-user@ip-172-31-18-39 html]$ sudo nginx -t
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
[ec2-user@ip-172-31-18-39 html]$ sudo systemctl restart nginx
[ec2-user@ip-172-31-18-39 html]$ sudo systemctl restart php-fpm
[ec2-user@ip-172-31-18-39 html]$ nano /usr/share/nginx/html/test.php
[ec2-user@ip-172-31-18-39 html]$
```

5. Website

○ Homepage



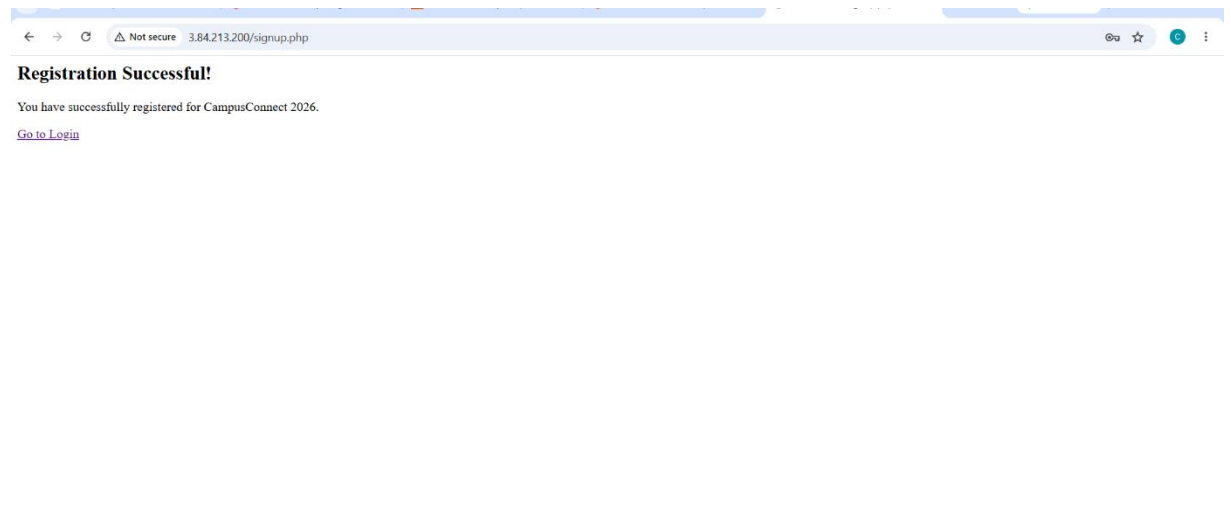
○ Registration form

The screenshot shows the 'Student Event Registration' form in a web browser window. The browser's address bar shows '3.84.213.200/register.php'. The form is centered on a light gray background and contains the following fields:

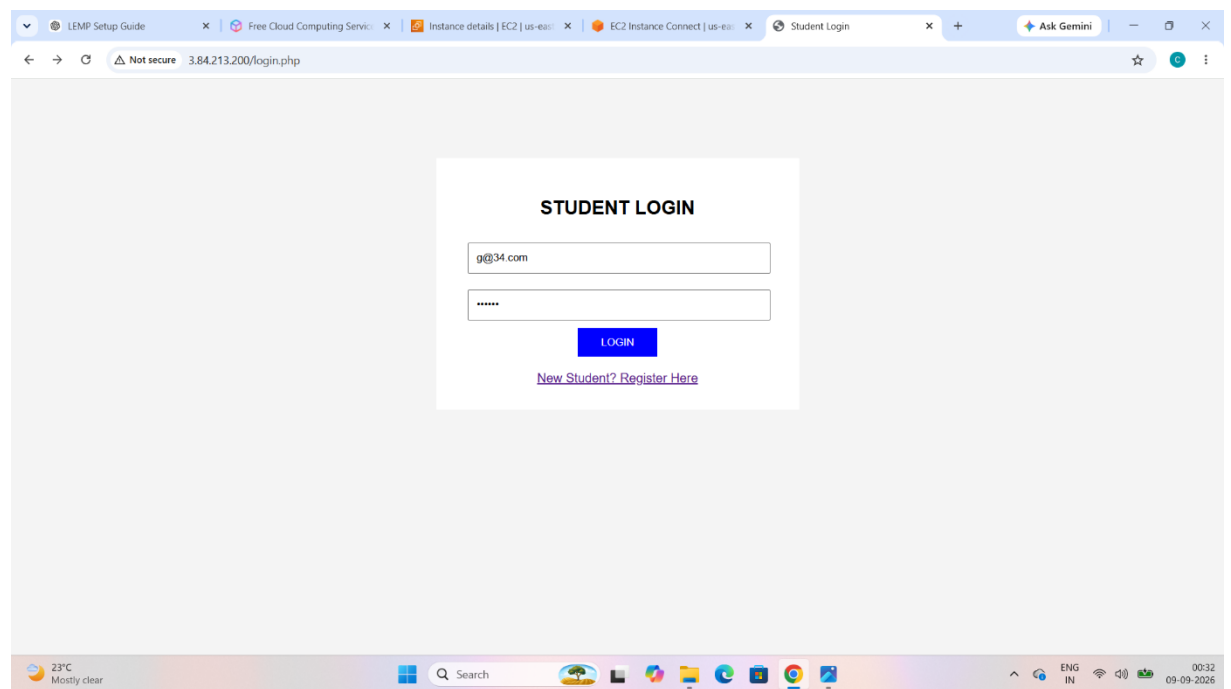
- Full Name**: Text input field containing 'naina'.
- Student ID**: Text input field containing '1'.
- Email**: Text input field containing 'g@34.com'.
- College Name**: Text input field containing 'DY patil'.
- Location**: Text input field containing 'pune'.
- Event**: Dropdown menu with 'Cultural Event' selected.
- Password**: Password input field with masked characters '*****'.
- REGISTER**: A green button at the bottom of the form.

The browser's taskbar at the bottom shows the system clock as 00:29 on 09-09-2026, along with various system icons and a search bar.

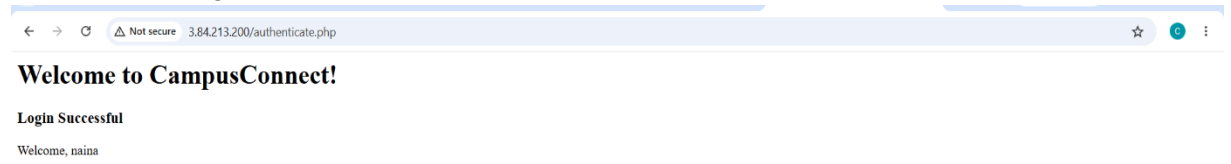
- Successful registration



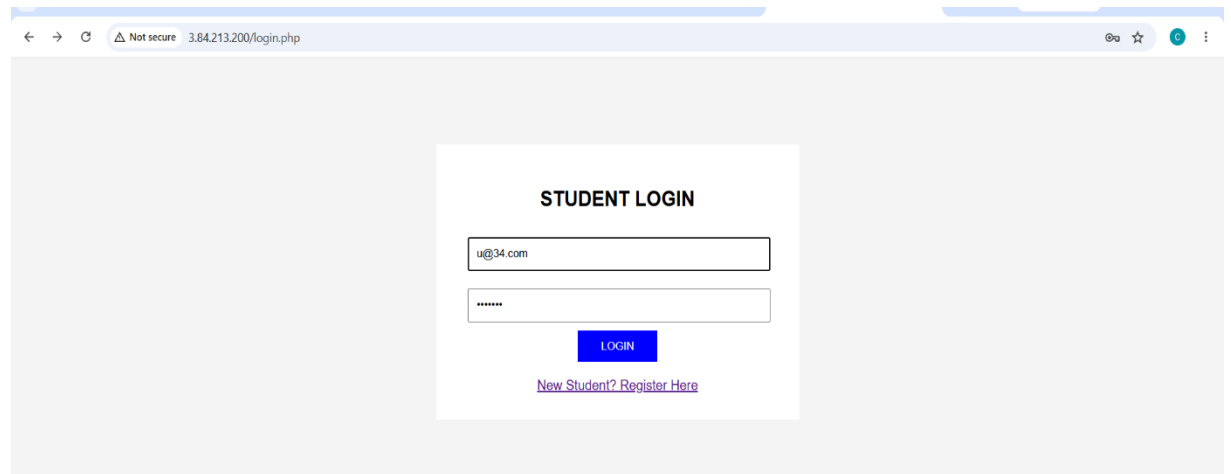
- Login form



○ Successful login



○ Invalid login test



6. Final Testing

- Service/port verification

The screenshot shows an AWS CloudShell terminal window with the following content:

```
[ec2-user@ip-172-31-18-39 ~]$ sudo ss -tulpn
Netid State Recv-Q Send-Q Local Address:Port Peer Address:Port Process
udp UNCONN 0 0 172.31.18.39%ens5:68 0.0.0.0:* users: (("systemd-network",pid=1713,fd=17))
udp UNCONN 0 0 127.0.0.1:323 0.0.0.0:* users: (("chronyd",pid=1919,fd=5))
udp UNCONN 0 0 [fe80::8ff:e6ff:fedc:7b21]:::ens5:546 [::]:* users: (("systemd-network",pid=1713,fd=19))
udp UNCONN 0 0 [::1]:323 [::]:* users: (("chronyd",pid=1919,fd=6))
tcp LISTEN 0 511 0.0.0.0:80 0.0.0.0:* users: (("nginx",pid=31405,fd=8), ("nginx",pid=31404,fd=8), ("nginx",pid=31403,fd=8))
tcp LISTEN 0 128 0.0.0.0:22 0.0.0.0:* users: (("sshd",pid=1890,fd=7))
tcp LISTEN 0 511 [::]:80 [::]:* users: (("nginx",pid=31405,fd=9), ("nginx",pid=31404,fd=9), ("nginx",pid=31403,fd=9))
tcp LISTEN 0 128 [::]:22 [::]:* users: (("sshd",pid=1890,fd=8))
tcp LISTEN 0 80 *:3306 *:3306 users: (("mariadb",pid=31726,fd=20))

[ec2-user@ip-172-31-18-39 ~]$ sudo ss -tulpn | grep :80
tcp LISTEN 0 511 0.0.0.0:80 0.0.0.0:* users: (("nginx",pid=31405,fd=8), ("nginx",pid=31404,fd=8), ("nginx",pid=31403,fd=8))
tcp LISTEN 0 511 [::]:80 [::]:* users: (("nginx",pid=31405,fd=9), ("nginx",pid=31404,fd=9), ("nginx",pid=31403,fd=9))

[ec2-user@ip-172-31-18-39 ~]$
```

Below the terminal output, a summary box for instance **i-01eaa8884d0e88985 (practical exam)** is shown, listing PublicIPs: 3.84.213.200 and PrivateIPs: 172.31.18.39.

The bottom of the screenshot shows the AWS CloudShell interface with a search bar, navigation icons, and a taskbar at the very bottom displaying system information like 23°C, 00:40, and 09-09-2026.